

AgriSentinel — SIH Hackathon 2026

Digital Farm Management Portal for Biosecurity in Pig & Poultry Farms

Problem Statement ID: SIH260487

Team Platform: AgriSentinel — Digital Farm Biosecurity Platform

Document Version: August 2026

Live Demo Links

Resource	URL
Website (Demo)	https://backend-farm-mdws.vercel.app
Backend API	https://agrisentinel-api.onrender.com
API Documentation	https://agrisentinel-api.onrender.com/docs
GitHub Repository	https://github.com/ullassallu28-eng/BACKEND-FARM-

1. Executive Summary

AgriSentinel is a full-stack **Digital Farm Biosecurity Portal** designed for **pig and poultry farms**. It provides a closed-loop system covering farm registration, biosecurity assessment, risk monitoring, incident reporting, veterinary verification, corrective actions, and government oversight through GIS mapping.

The platform supports three user roles — **Farmer**, **Veterinarian**, and **Government Officer** — each with a dedicated portal connected to a central PostgreSQL database via a FastAPI backend.

2. Problem Statement Alignment (SIH260487)

SIH Requirement	AgriSentinel Solution
Digital farm management for pig & poultry farms	Multi-farm portal with poultry, pig, and mixed farm types
Biosecurity monitoring & compliance	Biosecurity score, digital passport, daily checklist
Incident reporting & outbreak control	Incident form with photo evidence; vet verification workflow
Risk assessment & analytics	Rule-based risk engine with factors and score history
Role-based multi-stakeholder access	Farmer, Veterinarian, and Government Officer portals
Corrective actions & evidence tracking	Action list with evidence upload and verification
Regional / district oversight	Officer dashboard and GIS farm map
Closed-loop biosecurity response	Incident → Verify → Corrective Action → Score Update

3. Role-Based Features

3.1 Farmer Portal

- Farm dashboard with biosecurity score (0–100) and risk level (Safe / Caution / Critical)
- Multi-farm selection across registered poultry and pig farms
- Routine biosecurity verification checklist (sanitation, disinfection, visitor logs)
- Report farm health incidents with:
 - Incident category (mortality, respiratory, contamination, perimeter breach, etc.)
 - Animal species, batch, and number affected
 - Farm zone location, date/time, and symptom description
 - Photo and document evidence upload
- Recent farm incident log on dashboard
- Digital Biosecurity Passport with component scores and inspection history
- Risk analytics dashboard showing active risk factors
- Corrective actions list with compliance evidence submission
- Aarohi AI Biosecurity Advisor — contextual tips from live risk data
- Real-time notifications for incidents and verifications

3.2 Veterinarian Portal

- District Veterinary Verification Queue
- Review incoming incident reports with full farm and health details
- Inspect uploaded diagnostic evidence (photos/documents)
- Validate (verify) incidents — auto-generates corrective actions
- Request additional information from farmers
- Reject non-critical or false-positive reports
- Add veterinary inspection notes and comments
- Track workflow status: Reported → Under Review → Verified / Rejected

3.3 Government Officer Portal

- District field command overview
- Regional farm statistics and inspection priority list
- GIS farm map with location-based risk indicators
- Identify high-risk farms requiring field inspection
- Cross-farm biosecurity oversight for the district

4. Core Platform Modules

#	Module	Description	Status
1	Farm Registration & Profile	Farm ID, owner, type, location, animal population	✔ Live
2	Biosecurity Passport	Digital passport with scores, QR code, compliance status	✔ Live
3	Biosecurity Assessment	Component scores: hygiene, visitor control, quarantine, waste	✔ Live

4	Risk Calculation Engine	Rule-based scoring from incidents, mortality, sanitation	✓ Live
5	Incident Reporting	Form submission with multipart photo/document upload	✓ Live
6	Veterinary Verification	Validate, reject, or request info on reported incidents	✓ Live
7	Corrective Actions	Auto-generated after verification; evidence tracking	✓ Live
8	Checklist Management	Daily biosecurity verification tasks per farm	✓ Live
9	GIS Mapping	Geographic farm nodes with risk context	✓ Live
10	Notifications	Role-based alerts for incidents, verifications, risk updates	✓ Live
11	Authentication & Roles	JWT-based login for Farmer, Vet, Officer	✓ Live
12	Aarohi AI Advisor	Rule-based biosecurity guidance from risk data	✓ Live
13	Health Records API	Farm health record management endpoints	✓ Backend Ready
14	Spatial Risk API	Nearby farm risk and spatial analysis	✓ Backend Ready

5. Closed-Loop Biosecurity Workflow

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Farm Registration
↓
Biosecurity Assessment & Score Calculation
↓
Continuous Monitoring (Checklist + Risk Factors)
↓
Health Incident Reported (with Photo Evidence)
↓
Veterinary Verification Queue
↓
Validate / Reject / Request More Information
↓
Corrective Actions Auto-Generated
↓
Farmer Submits Compliance Evidence
↓
Biosecurity Score Updated
↓
Government Officer – GIS & District Overview

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6. Technology Stack

Layer	Technology
Frontend	React 19, TypeScript, Vite
Backend API	FastAPI (Python), 40+ REST endpoints
Database	PostgreSQL (Alembic migrations)
Authentication	JWT (JSON Web Tokens)
Website Hosting	Vercel
API & Database Hosting	Render Cloud
Version Control	GitHub (Monorepo: BACKEND-FARM-)

7. Team Responsibilities

Team Member	Responsibility
Backend / API Developer	FastAPI backend, deployment, API integration, workflows
Database Developer	PostgreSQL schema, migrations, seed data, real farm imports
AI / ML Developer	Image-based disease detection, medicine recommendations (planned)
Frontend Developer	React UI/UX integrated with live API
IoT Developer	Sensor telemetry integration (planned)

8. Future Enhancements (Roadmap)

Feature	Status
AI image diagnosis from uploaded farm photos	➡️ SOON Planned (AI Team)
Automated medicine & treatment recommendations	➡️ SOON Planned (AI Team)
Live IoT sensor feeds (temperature, mortality sensors)	➡️ SOON Planned (IoT Team)
Real Jharkhand farm database import	➡️ SOON In Progress (DB Team)
SMS / WhatsApp alerts to farmers	➡️ SOON Future Enhancement

9. Demo Script (2–3 Minutes for Judges)

- Farmer Role** — Show dashboard, biosecurity score, and daily checklist
- Report Incident** — Fill form, upload photo evidence, submit report
- Veterinarian Role** — Open verification queue, review evidence, validate incident
- Corrective Actions** — Show auto-generated actions for the farmer
- Government Officer** — Display GIS farm map and district overview
- Aarohi Advisor** — Click AI assistant for contextual biosecurity tips
- Biosecurity Passport** — Open digital passport with scores and history

10. Key Differentiators

- **Full-stack live deployment** — not a prototype; fully cloud-hosted demo
- **True multi-role workflow** — Farmer, Vet, and Officer portals share one database
- **Evidence-based incident reporting** — photo upload attached to veterinary review
- **Closed-loop design** — every incident flows through verification to corrective action
- **Scalable architecture** — modular API ready for AI, IoT, and real farm data integration
- **Aligned with SIH260487** — directly addresses pig & poultry biosecurity management

AgriSentinel — Protecting Farms, Securing Food Supply
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